

РБНФ №1 (опис синтаксису всіма допустимими засобами РБНФ)	РБНФ №2 (опис формальної граматики засобами РБНФ)	Формальна граматика	Формальна граматика з специфікацією lookahead у правилах для LL(2)-аналізатора	Перевірка РБНФ №1 за допомогою коду <i>(помістити у файл "EBNF_N1.h")</i> */	Перевірка РБНФ №2 за допомогою коду <i>(помістити у файл "EBNF_N2.h")</i> */	Перевірки прототипу LL(2)-синтаксичного аналізатора (спеціальна структура) та прототипу лексичного аналізатора (регулярні вирази) за допомогою коду. Лексеми для синтаксичного аналізатора обробляються лексичним аналізатором, тому синтаксичний аналізатор не аналізує їх повторно (як показано в РБНФ). <i>(помістити у файл "LexicaByRegExAndSyntaxByLL2prototype.h")</i> УВАГА: при копіюванні зажайте, щоб у кожному рядку після символу «\» не містилось жодних інших символів. */
		G = {N, T, P, S}	G = {N, T, P, S}			
		S → program_rule	S → program_rule			
		N = { program_name, value_type, array_specify, declaration_element, array_specify_optional, other_declaration_ident, declaration, other_declaration_ident__iteration, index_action, unary_operator, unary_operation, binary_operator, binary_action, left_expression, group_expression, index_action_optional, expression, binary_action__iteration, expression_or_cond_block_with_optional_assign, assign_to_right, assign_to_right_optional, if_expression, body_for_true, false_cond_block_without_else, body_for_false, cond_block, false_cond_block_without_else__iteration, body_for_false_optional, continue_while, break_while, statement_in_while_and_if_body, statement, block_statements_in_while_and_if_body, statement_in_while_and_if_body__iteration, while_cycle_head_expression, while_cycle, statements_or_block_statements, block_statements, input_rule, argument_for_input, output_rule, statement__iteration, expression_optional, program_rule, declaration_optional, non_zero_digit, digit__iteration, digit, unsigned_value, value, sign_optional, sign, ident, letter_in_upper_case, letter_in_lower_case, sign_plus, sign_minus }	N = { program_name, value_type, array_specify, declaration_element, array_specify_optional, other_declaration_ident, declaration, other_declaration_ident__iteration, index_action, unary_operator, unary_operation, binary_operator, binary_action, left_expression, group_expression, index_action_optional, expression, binary_action__iteration, expression_or_cond_block_with_optional_assign, assign_to_right, assign_to_right_optional, if_expression, body_for_true, false_cond_block_without_else, body_for_false, cond_block, false_cond_block_without_else__iteration, body_for_false_optional, continue_while, break_while, statement_in_while_and_if_body, statement, block_statements_in_while_and_if_body, statement_in_while_and_if_body__iteration, while_cycle_head_expression, while_cycle, statements_or_block_statements, block_statements, input_rule, argument_for_input, output_rule, statement__iteration, expression_optional, program_rule, declaration_optional, non_zero_digit, digit__iteration, digit, unsigned_value, value, sign_optional, sign, ident, letter_in_upper_case, letter_in_lower_case, sign_plus, sign_minus }	#define NONTERMINALS program_name, \ value_type, \ array_specify, \ declaration_element, \ \ other_declaration_ident, \ declaration, \ \ index_action, \ unary_operator, \ unary_operation, \ binary_operator, \ binary_action, \ left_expression, \ group_expression, \ \ expression, \ \ expression_or_cond_block_with_optional_assign, \ assign_to_right, \ \ if_expression, \ body_for_true, \ false_cond_block_without_else, \ body_for_false, \ cond_block, \ \ continue_while, \ break_while, \ statement_in_while_and_if_body, \ statement, \ block_statements_in_while_and_if_body, \ while_cycle_head_expression, \ while_cycle, \ statements_or_block_statements, \ block_statements, \ input_rule, \ argument_for_input, \ output_rule, \ \ program_rule, \ \ non_zero_digit, \ digit__iteration, \ digit, \ unsigned_value, \ value, \ \ sign, \ ident, \ letter_in_upper_case, \ letter_in_lower_case, \ sign_plus, \ sign_minus	#define NONTERMINALS program_name, \ value_type, \ array_specify, \ declaration_element, \ array_specify_optional, \ other_declaration_ident, \ declaration, \ other_declaration_ident__iteration, \ index_action, \ unary_operator, \ unary_operation, \ binary_operator, \ binary_action, \ left_expression, \ group_expression, \ index_action_optional, \ expression, \ binary_action__iteration, \ expression_or_cond_block_with_optional_assign, \ assign_to_right, \ assign_to_right_optional, \ if_expression, \ body_for_true, \ false_cond_block_without_else, \ body_for_false, \ cond_block, \ false_cond_block_without_else__iteration, \ body_for_false_optional, \ continue_while, \ break_while, \ statement_in_while_and_if_body, \ statement, \ block_statements_in_while_and_if_body, \ statement_in_while_and_if_body__iteration, \ while_cycle_head_expression, \ while_cycle, \ statements_or_block_statements, \ block_statements, \ input_rule, \ argument_for_input, \ output_rule, \ statement__iteration, \ expression_optional, \ program_rule, \ declaration_optional, \ non_zero_digit, \ digit__iteration, \ digit, \ unsigned_value, \ value, \ sign_optional, \ sign, \ ident, \ letter_in_upper_case, \ letter_in_lower_case, \ sign_plus, \ sign_minus	
		T = { "INTEGER16", " ,", "NOT", "AND", "OR", "==" , "!=", "<", ">", "+", "-", "*", "/", "DIV", "MOD", "(", 	T = { "INTEGER16", " ,", "NOT", "AND", "OR", "==" , "!=", "<", ">", "+", "-", "*", "/", "DIV", "MOD", "(", 	#define TOKENS \ tokenINTEGER16, \ tokenCOMMA, \ tokenNOT, \ tokenAND, \ tokenOR, \ tokenEQUAL, \ tokenNOTEQUAL, \ tokenLESS, \ tokenGREATER, \ tokenPLUS, \ tokenMINUS, \ tokenMUL, \ tokenDIV, \ tokenMOD, \ tokenGROUPEXPRESSIONBEGIN, \ 	#define TOKENS \ tokenINTEGER16, \ tokenCOMMA, \ tokenNOT, \ tokenAND, \ tokenOR, \ tokenEQUAL, \ tokenNOTEQUAL, \ tokenLESS, \ tokenGREATER, \ tokenPLUS, \ tokenMINUS, \ tokenMUL, \ tokenDIV, \ tokenMOD, \ tokenGROUPEXPRESSIONBEGIN, \ 	

[illegible]

				tokenBEGINBLOCK = "{" >> BOUNDARIES;	tokenBEGINBLOCK = "{" >> BOUNDARIES;	#define T_BEGIN_BLOCK_0 "{" #define T_BEGIN_BLOCK_1 "" #define T_BEGIN_BLOCK_2 "" #define T_BEGIN_BLOCK_3 ""
				tokenENDBLOCK = "}" >> BOUNDARIES;	tokenENDBLOCK = "}" >> BOUNDARIES;	#define T_END_BLOCK_0 "}" #define T_END_BLOCK_1 "" #define T_END_BLOCK_2 "" #define T_END_BLOCK_3 ""
				tokenSEMICOLON = ";" >> BOUNDARIES;	tokenSEMICOLON = ";" >> BOUNDARIES;	#define T_SEMICOLON_0 ";" #define T_SEMICOLON_1 "" #define T_SEMICOLON_2 "" #define T_SEMICOLON_3 ""
				tokenINTEGER16 = "INTEGER16" >> STRICT_BOUNDARIES;	tokenINTEGER16 = "INTEGER16" >> STRICT_BOUNDARIES;	#define T_DATA_TYPE_0 "INTEGER16" #define T_DATA_TYPE_1 "" #define T_DATA_TYPE_2 "" #define T_DATA_TYPE_3 ""
				tokenCOMMA = "," >> BOUNDARIES;	tokenCOMMA = "," >> BOUNDARIES;	#define T_COMA_0 "," #define T_COMA_1 "" #define T_COMA_2 "" #define T_COMA_3 ""
						#define T_BITWISE_NOT_0 "~" #define T_BITWISE_NOT_1 "" #define T_BITWISE_NOT_2 "" #define T_BITWISE_NOT_3 ""
				tokenNOT = "NOT" >> STRICT_BOUNDARIES;	tokenNOT = "NOT" >> STRICT_BOUNDARIES;	#define T_NOT_0 "NOT" #define T_NOT_1 "" #define T_NOT_2 "" #define T_NOT_3 ""
						#define T_BITWISE_AND_0 "&" #define T_BITWISE_AND_1 "" #define T_BITWISE_AND_2 "" #define T_BITWISE_AND_3 ""
				tokenAND = "AND" >> STRICT_BOUNDARIES;	tokenAND = "AND" >> STRICT_BOUNDARIES;	#define T_AND_0 "AND" #define T_AND_1 "" #define T_AND_2 "" #define T_AND_3 ""
						#define T_BITWISE_OR_0 " " #define T_BITWISE_OR_1 "" #define T_BITWISE_OR_2 "" #define T_BITWISE_OR_3 ""
				tokenOR = "OR" >> STRICT_BOUNDARIES;	tokenOR = "OR" >> STRICT_BOUNDARIES;	#define T_OR_0 "OR" #define T_OR_1 "" #define T_OR_2 "" #define T_OR_3 ""
				tokenEQUAL = "==" >> BOUNDARIES;	tokenEQUAL = "==" >> BOUNDARIES;	#define T_EQUAL_0 "==" #define T_EQUAL_1 "" #define T_EQUAL_2 "" #define T_EQUAL_3 ""
				tokenNOTEQUAL = "!=" >> BOUNDARIES;	tokenNOTEQUAL = "!=" >> BOUNDARIES;	#define T_NOT_EQUAL_0 "!=" #define T_NOT_EQUAL_1 "" #define T_NOT_EQUAL_2 "" #define T_NOT_EQUAL_3 ""
				tokenLESS = "<" >> BOUNDARIES;	tokenLESS = "<" >> BOUNDARIES;	#define T_LESS_0 "<" #define T_LESS_1 "" #define T_LESS_2 "" #define T_LESS_3 ""
				tokenGREATER = ">" >> BOUNDARIES;	tokenGREATER = ">" >> BOUNDARIES;	#define T_GREATER_0 ">" #define T_GREATER_1 "" #define T_GREATER_2 "" #define T_GREATER_3 ""
				tokenPLUS = "+" >> BOUNDARIES;	tokenPLUS = "+" >> BOUNDARIES;	#define T_ADD_0 "+" #define T_ADD_1 "" #define T_ADD_2 "" #define T_ADD_3 ""
				tokenMINUS = "-" >> BOUNDARIES;	tokenMINUS = "-" >> BOUNDARIES;	#define T_SUB_0 "-" #define T_SUB_1 "" #define T_SUB_2 "" #define T_SUB_3 ""
				tokenMUL = "*" >> BOUNDARIES;	tokenMUL = "*" >> BOUNDARIES;	#define T_MUL_0 "*" #define T_MUL_1 "" #define T_MUL_2 "" #define T_MUL_3 ""
				tokenDIV = "DIV" >> STRICT_BOUNDARIES;	tokenDIV = "DIV" >> STRICT_BOUNDARIES;	#define T_DIV_0 "DIV" #define T_DIV_1 "" #define T_DIV_2 "" #define T_DIV_3 ""
				tokenMOD = "MOD" >> STRICT_BOUNDARIES;	tokenMOD = "MOD" >> STRICT_BOUNDARIES;	#define T_MOD_0 "MOD" #define T_MOD_1 "" #define T_MOD_2 "" #define T_MOD_3 ""
				tokenLRASSIGN = "=:;" >> BOUNDARIES;	tokenLRASSIGN = "=:;" >> BOUNDARIES;	#define T_LRASSIGN_0 "=:;" #define T_LRASSIGN_1 "" #define T_LRASSIGN_2 "" #define T_LRASSIGN_3 ""
						#define T_THEN_BLOCK_0 "{" #define T_THEN_BLOCK_1 "" #define T_THEN_BLOCK_2 "" #define T_THEN_BLOCK_3 ""
				tokenELSE = "ELSE" >> STRICT_BOUNDARIES;	tokenELSE = "ELSE" >> STRICT_BOUNDARIES;	#define T_ELSE_BLOCK_0 "ELSE" #define T_ELSE_BLOCK_1 T_BEGIN_BLOCK_0 #define T_ELSE_BLOCK_2 "" #define T_ELSE_BLOCK_3 ""
				tokenIF = "IF" >> STRICT_BOUNDARIES;	tokenIF = "IF" >> STRICT_BOUNDARIES;	#define T_IF_0 "IF" #define T_IF_1 "" #define T_IF_2 "" #define T_IF_3 ""
						#define T_ELSE_IF_0 T_ELSE_BLOCK_0 #define T_ELSE_IF_1 T_IF_0 #define T_ELSE_IF_2 "" #define T_ELSE_IF_3 ""
				tokenWHILE = "WHILE" >> STRICT_BOUNDARIES;	tokenWHILE = "WHILE" >> STRICT_BOUNDARIES;	#define T_WHILE_0 "WHILE" #define T_WHILE_1 "" #define T_WHILE_2 "" #define T_WHILE_3 ""
				tokenCONTINUE = "CONTINUE" >> STRICT_BOUNDARIES;	tokenCONTINUE = "CONTINUE" >> STRICT_BOUNDARIES;	#define T_CONTINUE_WHILE_0 "CONTINUE" #define T_CONTINUE_WHILE_1 "" #define T_CONTINUE_WHILE_2 ""

					#define T_CONTINUE_WHILE_3 ""
				tokenBREAK = "BREAK" >> STRICT_BOUNDARIES;	#define T_EXIT_WHILE_0 "BREAK" #define T_EXIT_WHILE_1 "" #define T_EXIT_WHILE_2 "" #define T_EXIT_WHILE_3 ""
				tokenEXIT = "EXIT" >> STRICT_BOUNDARIES;	#define T_EXIT_0 "EXIT" #define T_EXIT_1 "" #define T_EXIT_2 "" #define T_EXIT_3 ""
				tokenGET = "GET" >> STRICT_BOUNDARIES;	#define T_INPUT_0 "GET" #define T_INPUT_1 "" #define T_INPUT_2 "" #define T_INPUT_3 ""
				tokenPUT = "PUT" >> STRICT_BOUNDARIES;	#define T_OUTPUT_0 "PUT" #define T_OUTPUT_1 "" #define T_OUTPUT_2 "" #define T_OUTPUT_3 ""
				tokenNAME = "NAME" >> STRICT_BOUNDARIES;	#define T_NAME_0 "NAME" #define T_NAME_1 "" #define T_NAME_2 "" #define T_NAME_3 ""
				tokenBODY = "BODY" >> STRICT_BOUNDARIES;	#define T_BODY_0 "BODY" #define T_BODY_1 "" #define T_BODY_2 "" #define T_BODY_3 ""
				tokenDATA = "DATA" >> STRICT_BOUNDARIES;	#define T_DATA_0 "DATA" #define T_DATA_1 "" #define T_DATA_2 "" #define T_DATA_3 ""
				tokenBEGIN = "BEGIN" >> STRICT_BOUNDARIES;	#define T_BEGIN_0 "BEGIN" #define T_BEGIN_1 "" #define T_BEGIN_2 "" #define T_BEGIN_3 ""
				tokenEND = "END" >> STRICT_BOUNDARIES;	#define T_END_0 "END" #define T_END_1 "" #define T_END_2 "" #define T_END_3 ""
					#define T_NULL_STATEMENT_0 "NULL" #define T_NULL_STATEMENT_1 "STATEMENT" #define T_NULL_STATEMENT_2 "" #define T_NULL_STATEMENT_3 ""
					#define GRAMMAR_LL2_2025 {\
program_name = ident;	program_name = ident;	program_name → ident	program_name(1: "ident_terminal") → ident	program_name = SAME_RULE(ident);	{ LA_IS, ("ident_terminal"), { "program_name", {\
value_type = "INTEGER16";	value_type = "INTEGER16";	value_type → "INTEGER16"	value_type(1: "INTEGER16") → "INTEGER16"	value_type = SAME_RULE(tokenINTEGER16);	{ LA_IS, (T_DATA_TYPE_0), { "value_type", {\
	array_specify = "[", unsigned_value, "]";	array_specify → "[" unsigned_value "]"	array_specify(1: "["") → "[" unsigned_value "]"		{ LA_IS, ("["), { "array_specify", {\
declaration_element = ident, array_specify__optional;	declaration_element = ident, array_specify__optional;	declaration_element → ident array_specify__optional	declaration_element(1: "ident_terminal") → ident array_specify__optional	declaration_element = ident >> -(tokenLEFTSQUAREBRACKETS >> unsigned_value >> tokenRIGHTSQUAREBRACKETS);	{ LA_IS, ("ident_terminal"), { "declaration_element", {\
	array_specify__optional = array_specify ε;	array_specify__optional → array_specify array_specify__optional → ε	array_specify__optional(1: "["") → array_specify array_specify__optional(1: !"["") → ε		{ LA_IS, ("["), { "array_specify__optional", {\
other_declaration_ident = " , ", declaration_element;	other_declaration_ident = " , ", declaration_element;	other_declaration_ident → " , ", declaration_element	other_declaration_ident(1: " , ") → " , ", declaration_element	other_declaration_ident = tokenCOMMA >> declaration_element;	{ LA_IS, (T_COMA_0), { "other_declaration_ident", {\
declaration = value_type , declaration_element , other_declaration_ident_iteration;	declaration = value_type , declaration_element , other_declaration_ident_iteration;	declaration → value_type declaration_element other_declaration_ident_iteration	declaration(1: "INTEGER16") → value_type declaration_element other_declaration_ident_iteration	declaration = value_type >> declaration_element >> *other_declaration_ident;	{ LA_IS, (T_DATA_TYPE_0), { "declaration", {\
	other_declaration_ident_iteration = other_declaration_ident, other_declaration_ident_iteration ε;	other_declaration_ident_iteration → other_declaration_ident other_declaration_ident_iteration false_cond_block_without_else_iteration → ε	other_declaration_ident_iteration(1: " , ") → other_declaration_ident other_declaration_ident_iteration false_cond_block_without_else_iteration(1: !" , ") → ε		{ LA_IS, (T_COMA_0), {\
index_action = "[", expression, "]";	index_action = "[", expression, "]";	index_action → "[" expression "]"	index_action(1: "["") → "[" expression "]"	index_action = tokenLEFTSQUAREBRACKETS >> expression >> tokenRIGHTSQUAREBRACKETS;	{ LA_IS, ("["), { "index_action", {\
unary_operator = "NOT";	unary_operator = "NOT";	unary_operator → "NOT"	unary_operator(1: "NOT") → "NOT"	unary_operator = SAME_RULE(tokenNOT);	{ LA_IS, (T_NOT_0), { "unary_operator", {\
unary_operation = unary_operator , expression;	unary_operation = unary_operator , expression;	unary_operation → unary_operator expression	unary_operation(1: "NOT") → unary_operator expression	unary_operation = unary_operator >> expression;	{ LA_IS, (T_NOT_0), { "unary_operation", {\
binary_operator = "AND" "OR" "==" "!=" "<" ">" "+" "-" "*" "DIV" "MOD";	binary_operator = "AND" "OR" "==" "!=" "<" ">" "+" "-" "*" "DIV" "MOD";	binary_operator → "AND" binary_operator → "OR" binary_operator → "==" binary_operator → "!=" binary_operator → "<" binary_operator → ">" binary_operator → "+" binary_operator → "-" binary_operator → "*" binary_operator → "DIV" binary_operator → "MOD"	binary_operator(1: "AND") → "AND" binary_operator(1: "OR") → "OR" binary_operator(1: "==") → "==" binary_operator(1: "!=") → "!=" binary_operator(1: "<") → "<" binary_operator(1: ">") → ">" binary_operator(1: "+") → "+" binary_operator(1: "-") → "-" binary_operator(1: "**") → "**" binary_operator(1: "DIV") → "DIV" binary_operator(1: "MOD") → "MOD"	binary_operator = tokenAND tokenOR tokenEQUAL tokenNOTEQUAL tokenLESS tokenGREATER tokenPLUS tokenMINUS tokenMUL tokenDIV tokenMOD;	{ LA_IS, (T_AND_0), { "binary_operator", {\

						{ LA_IS, { T_SUB_0 }, { "binary_operator", {\ { LA_IS, { "" }, 1, { T_SUB_0 }}\ }};\ { LA_IS, { T_MUL_0 }, { "binary_operator", {\ { LA_IS, { "" }, 1, { T_MUL_0 }}\ }};\ { LA_IS, { T_DIV_0 }, { "binary_operator", {\ { LA_IS, { "" }, 1, { T_DIV_0 }}\ }};\ { LA_IS, { T_MOD_0 }, { "binary_operator", {\ { LA_IS, { "" }, 1, { T_MOD_0 }}\ }};\ }
binary_action = binary_operator , expression;	binary_action = binary_operator , expression;	binary_action → binary_operator expression	binary_action(1: "AND", "OR", "=", "!=", "<", ">", "+", "-", "*", "DIV", "MOD") → binary_operator expression	binary_action = binary_operator >> expression;	binary_action = binary_operator >> expression;	{ LA_IS, { T_AND_0, T_OR_0, T_EQUAL_0, T_NOT_EQUAL_0, T_LESS_0, T_GREATER_0, T_ADD_0, T_SUB_0, T_MUL_0, T_DIV_0, T_MOD_0 }, {\ "binary_action", {\ { LA_IS, { "" }, 2, { "binary_operator", "expression" }}\ }};\ }
left_expression = group_expression unary_operation cond_block value ident , [index_action];	left_expression = group_expression unary_operation cond_block value ident , index_action__optional;	left_expression → group_expression left_expression → unary_operation left_expression → cond_block left_expression → value left_expression → ident , index_action__optional	left_expression(1: "(" → group_expression left_expression(1: "NOT") → unary_operation left_expression(1: "IF") → cond_block left_expression(1: "0", "1", "2", "3", "4", "5", "6", "7", "8", "9") → value left_expression(1: "+", "-", "0", "1", "2", "3", "4", "5", "6", "7", "8", "9") → value left_expression(1: "_") → ident , index_action__optional	left_expression = group_expression unary_operation cond_block value ident >> -index_action;	left_expression = group_expression unary_operation cond_block value ident >> index_action__optional;	{LA_IS, { "(" }, { "left_expression", {\ { LA_IS, { "" }, 1, { "group_expression" }}\ }};\ {LA_IS, { T_NOT_0 }, { "left_expression", {\ { LA_IS, { "" }, 1, { "unary_operation" }}\ }};\ {LA_IS, { T_IF_0 }, { "left_expression", {\ { LA_IS, { "" }, 1, { "cond_block" }}\ }};\ {LA_IS, { "unsigned_value_terminal" }, { "left_expression", {\ "left_expression", {\ { LA_IS, { "" }, 1, { "value" }}\ }};\ {LA_IS, { T_ADD_0, T_SUB_0 }, { "left_expression", {\ {LA_IS, { "unsigned_value_terminal" }, 1, { "value" }};\ /*{LA_NOT, { "unsigned_value_terminal" }, 1, { "unary_operation" }}*\ }};\ {LA_IS, { "ident_terminal" }, { "left_expression", {\ { LA_IS, { "" }, 2, { "ident", "index_action__optional" }}\ }};\ }
	index_action__optional = index_action ε;	index_action__optional → index_action index_action__optional → ε	index_action__optional(1: "(" → index_action index_action__optional(1: "!=") → ε		index_action__optional = index_action "";	{LA_IS, { "(" }, { "index_action__optional", {\ { LA_IS, { "" }, 1, { "index_action" }}\ }};\ {LA_NOT, { "(" }, { "index_action__optional", {\ { LA_IS, { "" }, 0, { "" }}\ }};\ }
expression = left_expression , {binary_action};	expression = left_expression , binary_action__iteration;	expression → left_expression binary_action__iteration	expression(1: "(" , "NOT", "+", "-", "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "IF") → left_expression binary_action__iteration	expression = left_expression >> *binary_action;	expression = left_expression >> binary_action__iteration;	{LA_IS, { "(" , T_NOT_0, T_ADD_0, T_SUB_0, "ident_terminal", "unsigned_value_terminal", T_IF_0 }, {\ "expression", {\ { LA_IS, { "" }, 2, { "left_expression", "binary_action__iteration" }}\ }};\ }
	binary_action__iteration = binary_action, binary_action__iteration ε;	binary_action__iteration → binary_action binary_action__iteration binary_action__iteration → ε	binary_action__iteration(1: "AND", "OR", "=", "!=", "<", ">", "+", "-", "*", "DIV", "MOD") → binary_action binary_action__iteration binary_action__iteration(1: "!"AND", "!"OR", "!=" , "!=" , "!"<", "!">", "!"+" , "!"-" , "!"*" , "!"DIV", "!"MOD") → ε		binary_action__iteration = binary_action >> binary_action__iteration "";	{LA_IS, { T_AND_0, T_OR_0, T_EQUAL_0, T_NOT_EQUAL_0, T_LESS_0, T_GREATER_0, T_ADD_0, T_SUB_0, T_MUL_0, T_DIV_0, T_MOD_0 }, {\ "binary_action__iteration", {\ { LA_IS, { "" }, 2, { "binary_action", "binary_action__iteration" }}\ }};\ {LA_NOT, { T_AND_0, T_OR_0, T_EQUAL_0, T_NOT_EQUAL_0, T_LESS_0, T_GREATER_0, T_ADD_0, T_SUB_0, T_MUL_0, T_DIV_0, T_MOD_0 }, {\ "binary_action__iteration", {\ { LA_IS, { "" }, 0, { "" }}\ }};\ }
group_expression = "(" , expression , ")";	group_expression = "(" , expression , ")";	group_expression → "(" expression ")"	group_expression(1: "(" → "(" expression ")"	group_expression = tokenGROUPEXPRESSSIONBEGIN >> expression >> tokenGROUPEXPRESSSIONEND;	group_expression = tokenGROUPEXPRESSSIONBEGIN >> expression >> tokenGROUPEXPRESSSIONEND;	{LA_IS, { "(" }, { "group_expression", {\ { LA_IS, { "" }, 3, { "(" , "expression", ")" }}\ }};\ }
expression_or_cond_block__with_optional_assign = expression , ["=" , ident , [index_action]];	expression_or_cond_block__with_optional_assign = expression , assign_to_right__optional;	expression_or_cond_block__with_optional_assign → expression assign_to_right__optional	expression_or_cond_block__with_optional_assign(1: "(" , "NOT", "+", "-", "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "IF") → expression assign_to_right__optional	expression_or_cond_block__with_optional_assign = expression >> -(tokenLRASSIGN >> ident >> -index_action);	expression_or_cond_block__with_optional_assign = expression >> assign_to_right__optional;	{LA_IS, { "(" , T_NOT_0, T_ADD_0, T_SUB_0, "ident_terminal", "unsigned_value_terminal", T_IF_0 }, {\ "expression_or_cond_block__with_optional_assign", {\ { LA_IS, { "" }, 2, { "expression", "assign_to_right__optional" }}\ }};\ }
	assign_to_right = "=", ident , index_action__optional;	assign_to_right → "=" ident index_action__optional	assign_to_right(1: "=" → "=" ident index_action__optional		assign_to_right = tokenLRASSIGN >> ident >> index_action__optional;	{LA_IS, { T_LRASSIGN_0 }, {\ "assign_to_right", {\ { LA_IS, { "" }, 3, { T_LRASSIGN_0, "ident", "index_action__optional" }}\ }};\ }
	assign_to_right__optional = assign_to_right ε;	assign_to_right__optional → assign_to_right assign_to_right__optional → ε;	assign_to_right__optional(1: "=" → assign_to_right assign_to_right__optional(1: "!=") → ε;		assign_to_right__optional = assign_to_right "";	{ LA_IS, { T_LRASSIGN_0 }, {\ "assign_to_right__optional", {\ { LA_IS, { "" }, 1, { "assign_to_right" }}\ }};\ { LA_NOT, { T_LRASSIGN_0 }, {\ "assign_to_right__optional", {\ { LA_IS, { "" }, 0, { "" }}\ }};\ }
if_expression = expression;	if_expression = expression;	if_expression → expression	if_expression(1: "(" , "NOT", "+", "-", "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "IF") → expression	if_expression = SAME_RULE(expression);	if_expression = SAME_RULE(expression);	{LA_IS, { "(" , T_NOT_0, T_ADD_0, T_SUB_0, "ident_terminal", "unsigned_value_terminal", T_IF_0 }, {\ "if_expression", {\ { LA_IS, { "" }, 1, { "expression" }}\ }};\ }
body_for_true = block_statements_in_while_and_if_body;	body_for_true = block_statements_in_while_and_if_body;	body_for_true → block_statements_in_while_and_if_body	body_for_true(1: "(" → block_statements_in_while_and_if_body	body_for_true = SAME_RULE(block_statements_in_while_and_if_body);	body_for_true = SAME_RULE(block_statements_in_while_and_if_body);	{LA_IS, { T_BEGIN_BLOCK_0 }, {\ "body_for_true", {\ { LA_IS, { "" }, 1, { "block_statements_in_while_and_if_body" }}\ }};\ }
false_cond_block_without_else = "ELSE" , "IF" , if_expression , body_for_true;	false_cond_block_without_else = "ELSE" , "IF" , if_expression , body_for_true;	false_cond_block_without_else → "ELSE" "IF" if_expression body_for_true	false_cond_block_without_else(1: "ELSE") → "ELSE" "IF" if_expression body_for_true	false_cond_block_without_else = tokenELSE >> tokenIF >> if_expression >> body_for_true;	false_cond_block_without_else = tokenELSE >> tokenIF >> if_expression >> body_for_true;	{LA_IS, { T_ELSE_IF_0 }, {\ "false_cond_block_without_else", {\ { LA_IS, { "" }, 4, { T_ELSE_IF_0, T_ELSE_IF_1, "if_expression", "body_for_true" }}\ }};\ }
body_for_false = "ELSE" , block_statements_in_while_and_if_body;	body_for_false = "ELSE" , block_statements_in_while_and_if_body;	body_for_false → "ELSE" block_statements_in_while_and_if_body	body_for_false(1: "ELSE") → "ELSE" block_statements_in_while_and_if_body	body_for_false = tokenELSE >> block_statements_in_while_and_if_body;	body_for_false = tokenELSE >> block_statements_in_while_and_if_body;	{LA_IS, { T_ELSE_BLOCK_0 }, {\ "body_for_false", {\ { LA_IS, { "" }, 2, { T_ELSE_BLOCK_0, "block_statements" }}\ }};\ }
cond_block = "IF" , if_expression , body_for_true, false_cond_block_without_else__iteration , body_for_false__optional;	cond_block = "IF" , if_expression , body_for_true, false_cond_block_without_else__iteration , body_for_false__optional;	cond_block → "IF" if_expression body_for_true false_cond_block_without_else__iteration body_for_false__optional	cond_block(1: "IF") → "IF" if_expression body_for_true false_cond_block_without_else__iteration body_for_false__optional	cond_block = tokenIF >> if_expression >> body_for_true >> *false_cond_block_without_else >> -body_for_false;	cond_block = tokenIF >> if_expression >> body_for_true >> false_cond_block_without_else__iteration >> body_for_false__optional;	{LA_IS, { T_IF_0 }, {\ "cond_block", {\ { LA_IS, { "" }, 5, { T_IF_0, "if_expression", "body_for_true", "false_cond_block_without_else__iteration", "body_for_false__optional" }}\ }};\ }

						}}}\
	false_cond_block_without_else__iteration = false_cond_block_without_else, false_cond_block_without_else__iteration ε;	false_cond_block_without_else__iteration → false_cond_block_without_else false_cond_block_without_else__iteration false_cond_block_without_else__iteration → ε	false_cond_block_without_else__iteration(1: "ELSE"; 2: "IF") → false_cond_block_without_else false_cond_block_without_else__iteration false_cond_block_without_else__iteration(1: "ELSE"; 2: !"IF") → ε false_cond_block_without_else__iteration(1: !"ELSE") → ε		false_cond_block_without_else__iteration = false_cond_block_without_else >> false_cond_block_without_else__iteration "";	{LA_IS, {T_ELSE_IF_0}, { "false_cond_block_without_else__iteration", {\n {LA_IS, {T_ELSE_IF_1}, 2, {\n "false_cond_block_without_else", "false_cond_block_without_else__iteration" }}\n {LA_NOT, {T_ELSE_IF_1}, 0, { "" }}\n }}}\
		body_for_false__optional = body_for_false ε;	body_for_false__optional → body_for_false body_for_false__optional → ε		body_for_false__optional = body_for_false "";	{LA_IS, {T_ELSE_BLOCK_0}, { "body_for_false__optional", {\n {LA_IS, {""}, 1, { "body_for_false" }}\n }}}\
		continue_while = "CONTINUE";	continue_while → "CONTINUE"	continue_while(1: "CONTINUE") → "CONTINUE"	continue_while = SAME_RULE(tokenCONTINUE);	{LA_IS, {T_CONTINUE_WHILE_0}, {"continue_while", {\n {LA_IS, {""}, 1, {T_CONTINUE_WHILE_0 }}\n }}}\
				continue_while = SAME_RULE(tokenCONTINUE);		
	break_while = "BREAK";	break_while → "BREAK"	break_while(1: "BREAK") → "BREAK"	break_while = SAME_RULE(tokenBREAK);	break_while = SAME_RULE(tokenBREAK);	{LA_IS, {T_EXIT_WHILE_0}, {"break_while", {\n {LA_IS, {""}, 1, {T_EXIT_WHILE_0 }}\n }}}\
statement_in_while_and_if_body = statement "CONTINUE" "BREAK";	statement_in_while_and_if_body = statement continue_while break_while;	statement_in_while_and_if_body → statement statement_in_while_and_if_body → continue_while statement_in_while_and_if_body → break_while	statement_in_while_and_if_body(1: " ", "(", "NOT", "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "4", "-", "IF", "WHILE", "GET", "PUT", ",") → statement statement_in_while_and_if_body(1: "CONTINUE") → continue_while statement_in_while_and_if_body(1: "BREAK") → break_while	statement_in_while_and_if_body = statement continue_while break_while;	statement_in_while_and_if_body = statement continue_while break_while;	{LA_IS, { "ident_terminal", "(", T_NOT_0, "unsigned_value_terminal", T_ADD_0, T_SUB_0, T_IF_0, T_WHILE_0, T_INPUT_0, T_OUTPUT_0, T_SEMICOLON_0 }, {\n "statement_in_while_and_if_body", {\n {LA_IS, {""}, 1, { "statement" }}\n }}}\
block_statements_in_while_and_if_body = "(" , {statement_in_while_and_if_body} , ")";	block_statements_in_while_and_if_body = "(" , statement_in_while_and_if_body__iteration , ")";	block_statements_in_while_and_if_body → "(" statement_in_while_and_if_body__iteration ")"	block_statements_in_while_and_if_body(1: "(") → "(" statement_in_while_and_if_body__iteration ")"	block_statements_in_while_and_if_body = tokenBEGINBLOCK >> *statement_in_while_and_if_body >> tokenENDBLOCK;	block_statements_in_while_and_if_body = tokenBEGINBLOCK >> statement_in_while_and_if_body__iteration >> tokenENDBLOCK;	{LA_IS, {T_BEGIN_BLOCK_0}, {\n "block_statements_in_while_and_if_body", {\n {LA_IS, {""}, 3, {T_BEGIN_BLOCK_0, "statement_in_while_and_if_body__iteration", T_END_BLOCK_0 }}\n }}}\
	statement_in_while_and_if_body__iteration = statement_in_while_and_if_body , statement_in_while_and_if_body__iteration ε;	statement_in_while_and_if_body__iteration → statement_in_while_and_if_body statement_in_while_and_if_body__iteration statement_in_while_and_if_body__iteration → ε	statement_in_while_and_if_body__iteration(1: " ", "(", "NOT", "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "4", "-", "IF", "WHILE", "GET", "PUT", ",", "CONTINUE", ";") → statement_in_while_and_if_body__iteration statement_in_while_and_if_body__iteration statement_in_while_and_if_body__iteration(1: !" ", !"(", !"NOT", !"0", !"1", !"2", !"3", !"4", !"5", !"6", !"7", !"8", !"9", !"4", !" ", !"IF", !"WHILE", !"GET", !"PUT", !" ", !"CONTINUE", !"BREAK") → ε		statement_in_while_and_if_body__iteration = statement_in_while_and_if_body >> statement_in_while_and_if_body__iteration "";	{LA_IS, { "ident_terminal", "(", T_NOT_0, "unsigned_value_terminal", T_ADD_0, T_SUB_0, T_IF_0, T_WHILE_0, T_INPUT_0, T_OUTPUT_0, T_SEMICOLON_0, T_CONTINUE_WHILE_0, T_EXIT_WHILE_0 }, {\n "statement_in_while_and_if_body__iteration", {\n {LA_IS, {""}, 2, { "statement_in_while_and_if_body", "statement_in_while_and_if_body__iteration" }}\n }}}\
while_cycle_head_expression = expression;	while_cycle_head_expression = expression;	while_cycle_head_expression → expression	while_cycle_head_expression(1: "(", "NOT", "+", "-", " ", "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "IF") → expression	while_cycle_head_expression = SAME_RULE(expression);	while_cycle_head_expression = SAME_RULE(expression);	{LA_IS, { "(", T_NOT_0, T_ADD_0, T_SUB_0, "ident_terminal", "unsigned_value_terminal", T_IF_0 , { "while_cycle_head_expression", {\n {LA_IS, {""}, 1, { "expression" }}\n }}}\
while_cycle = "WHILE" , while_cycle_head_expression , block_statements_in_while_and_if_body;	while_cycle = "WHILE" , while_cycle_head_expression , block_statements_in_while_and_if_body;	while_cycle → "WHILE" while_cycle_head_expression block_statements_in_while_and_if_body	while_cycle(1: "WHILE") → "WHILE" while_cycle_head_expression block_statements_in_while_and_if_body	while_cycle = tokenWHILE >> while_cycle_head_expression >> block_statements_in_while_and_if_body;	while_cycle = tokenWHILE >> while_cycle_head_expression >> block_statements_in_while_and_if_body;	{LA_IS, {T_WHILE_0}, {"while_cycle", {\n {LA_IS, {""}, 3, {T_WHILE_0, "while_cycle_head_expression", "block_statements_in_while_and_if_body" }}\n }}}\
	statements__or__block_statements = statement__iteration block_statements;	statements__or__block_statements → statement__iteration statements__or__block_statements → block_statements	statements__or__block_statements(1: " ", "(", "NOT", "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "+", "-", "IF", "WHILE", "GET", "PUT", ",") → statement__iteration statements__or__block_statements(1: "(") → block_statements		statements__or__block_statements = statement__iteration block_statements;	{LA_IS, { "ident_terminal", "(", T_NOT_0, "unsigned_value_terminal", T_ADD_0, T_SUB_0, T_IF_0, T_WHILE_0, T_INPUT_0, T_OUTPUT_0, T_SEMICOLON_0 }, {\n "statements__or__block_statements", {\n {LA_IS, {""}, 1, { "statement__iteration" }}\n }}}\
input_rule = "GET" , (ident , [index_action] "(" , ident , [index_action] , ")");	input_rule = "GET" , argument_for_input;	input_rule → "GET" argument_for_input	input_rule(1: "GET") → "GET" argument_for_input	input_rule = tokenGET >> {ident >> -index_action tokenGROUPEXPRESSIONBEGIN >> ident >> -index_action >> tokenGROUPEXPRESSIONEND};	input_rule = tokenGET >> argument_for_input;	{LA_IS, {T_INPUT_0}, {"input_rule", {\n {LA_IS, {""}, 2, {T_INPUT_0, "argument_for_input" }}\n }}}\
	argument_for_input = ident , index_action__optional; argument_for_input = "(" , "ident", "index_action__optional", ")";	argument_for_input → ident index_action__optional argument_for_input → "(" "ident" "index_action__optional" ")"	argument_for_input(1: " ") → ident index_action__optional argument_for_input(1: "(") → "(" "ident" "index_action__optional" ")"		argument_for_input = ident >> index_action__optional tokenGROUPEXPRESSIONBEGIN >> ident >> index_action__optional >> tokenGROUPEXPRESSIONEND;	{LA_IS, { " ident_terminal " }, {"argument_for_input", {\n {LA_IS, {""}, 2, { "ident", "index_action__optional" }}\n }}}\
output_rule = "PUT" , expression;	output_rule = "PUT" , expression;	output_rule → "PUT" expression	output(1: "PUT") → "PUT" expression	output_rule = tokenPUT >> expression;	output_rule = tokenPUT >> expression;	{LA_IS, {T_OUTPUT_0}, {"output_rule", {\n {LA_IS, {""}, 2, {T_OUTPUT_0, "expression" }}\n }}}\
statement = expression_or_cond_block__with_optional_assign while_cycle input_rule output_rule ";" ;	statement = expression_or_cond_block__with_optional_assign while_cycle input_rule output_rule ";" ;	statement → expression_or_cond_block__with_optional_assign statement → while_cycle statement → input_rule statement → output_rule statement → ";" ;	statement(1: " ", "2: !" ;") → expression_or_cond_block__with_optional_assign statement(1: "(", "NOT", "+", "-", "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "IF") → expression_or_cond_block__with_optional_assign statement(1: "WHILE") → while_cycle statement(1: "GET") → input_rule statement(1: "PUT") → output_rule statement(1: ";", "-") → ";" ;	statement = expression_or_cond_block__with_optional_assign while_cycle input_rule output_rule tokenSEMICOLON;	statement = expression_or_cond_block__with_optional_assign while_cycle input_rule output_rule tokenSEMICOLON;	{LA_IS, { "(", T_NOT_0, " ident_terminal ", "unsigned_value_terminal", T_ADD_0, T_SUB_0, T_IF_0 }, { "statement", {\n {LA_IS, {""}, 1, {"expression_or_cond_block__with_optional_assign" }}\n }}}\
statement = expression_or_cond_block__with_optional_assign while_cycle input_rule output_rule ";" ;				statement = expression_or_cond_block__with_optional_assign while_cycle input_rule output_rule tokenSEMICOLON;		{LA_IS, {T_WHILE_0}, {"statement", {\n {LA_IS, {""}, 1, { "while_cycle" }}\n }}}\

						{LA_IS, { T_INPUT_0 }, { "statement", {\ {LA_IS, {""}, 1, {"input_rule"}}}\ }}}\ {LA_IS, { T_OUTPUT_0 }, { "statement", {\ {LA_IS, {""}, 1, {"output_rule"}}}\ }}}\ {LA_IS, { T_SEMICOLON_0 }, { "statement", {\ {LA_IS, {""}, 1, {";"}}}\ }}}\
	statement__iteration = statement, statement__iteration ε;	statement__iteration → statement statement__iteration statement__iteration → ε	statement__iteration(1: !"_, "(", !"NOT", !"0", !"1", !"2", !"3", !"4", !"5", !"6", !"7", !"8", !"9", !"+", !"-", !"IF", !"WHILE", !"PUT", !";") → statement statement__iteration statement__iteration(1: !"_, "(", !"NOT", !"0", !"1", !"2", !"3", !"4", !"5", !"6", !"7", !"8", !"9", !"+", !"-", !"IF", !"WHILE", !"GET", !"PUT", !";") → ε		statement__iteration = statement >> statement__iteration "";	{ LA_IS, { "ident_terminal", "(", T_NOT_0, "unsigned_value_terminal", T_ADD_0, T_SUB_0, T_IF_0, T_WHILE_0, T_INPUT_0, T_OUTPUT_0, T_SEMICOLON_0 }, { "statement__iteration", {\ { LA_IS, {""}, 2, { "statement", "statement__iteration" }\ }}}\ }}}\ { LA_NOT, { "ident_terminal", "(", T_NOT_0, "unsigned_value_terminal", T_ADD_0, T_SUB_0, T_IF_0, T_WHILE_0, T_INPUT_0, T_OUTPUT_0, T_SEMICOLON_0 }, { "statement__iteration", {\ { LA_IS, {""}, 0, { "" } }\ }}}\
block_statements = "{" , {statement} , "}";	block_statements = "{" , statement__iteration , "}";	block_statements → "{" statement__iteration "}"	block_statements(1: !"(" → "{" statement__iteration "}"	block_statements = tokenBEGINBLOCK >> tokenENDBLOCK;	block_statements = tokenBEGINBLOCK >> statement__iteration >> tokenENDBLOCK;	{ LA_IS, { T_BEGIN_BLOCK_0 }, { "block_statements", {\ { LA_IS, {""}, 3, { T_BEGIN_BLOCK_0, "statement__iteration", T_END_BLOCK_0 } }\ }}}\
	expression__optional = expression "";	expression__optional → expression expression__optional → ε	expression__optional(1: !"(", !"NOT", !"+", !"-", !"0", !"1", !"2", !"3", !"4", !"5", !"6", !"7", !"8", !"9", !"IF") → expression expression__optional(1: !"(", !"NOT", !"+", !"-", !"_, !"0", !"1", !"2", !"3", !"4", !"5", !"6", !"7", !"8", !"9", !"IF") → ε		expression__optional = expression "";	{LA_IS, { "(", T_NOT_0, T_ADD_0, T_SUB_0, "ident_terminal", "unsigned_value_terminal", T_IF_0 , { "expression__optional", {\ {LA_IS, {""}, 1, { "expression" } }\ }}}\ }}}\ {LA_NOT, { "(", T_NOT_0, T_ADD_0, T_SUB_0, "ident_terminal", "unsigned_value_terminal", T_IF_0 , { "expression__optional", {\ {LA_IS, {""}, 0, { "" } }\ }}}\ }}}\
program_rule = "NAME" , program_name , "+", "BODY", "DATA", [declaration__optional , "+", statement__iteration , "END";	program_rule = "NAME" , program_name , "+", "BODY", "DATA", declaration__optional , "+", statement__iteration , "END";	program_rule → "NAME" program_name "+", "BODY" "DATA" declaration__optional "+", statement__iteration "END"	program_rule(1: "NAME") → "NAME" program_name "+", "BODY" "DATA" declaration__optional "+", statement__iteration "END"	program_rule = BOUNDARIES >> tokenNAME >> program_name >> tokenSEMICOLON >> tokenBODY >> tokenDATA >> (- declaration) >> tokenSEMICOLON >> *statement >> tokenEND;	program_rule = BOUNDARIES >> tokenNAME >> program_name >> tokenSEMICOLON >> tokenBODY >> tokenDATA >> declaration__optional >> tokenSEMICOLON >> statement__iteration >> tokenEND;	{ LA_IS, { T_NAME_0 }, { "program_rule", {\ { LA_IS, {""}, 9, { T_NAME_0, "program_name", T_SEMICOLON_0, T_BODY_0, T_DATA_0, "declaration__optional", T_SEMICOLON_0, "statement__iteration", T_END_0 } }\ }}}\
	declaration__optional = declaration "";	declaration__optional → declaration declaration__optional → ε	declaration__optional(1: "INTEGER16") → declaration declaration__optional(1: !"INTEGER16") → ε		declaration__optional = declaration "";	{ LA_IS, { T_DATA_TYPE_0 }, { "declaration__optional", {\ { LA_IS, {""}, 1, { "declaration" } }\ }}}\ { LA_NOT, { T_DATA_TYPE_0 }, { "declaration__optional", {\ { LA_IS, {""}, 0, { "" } }\ }}}\ }}}\
value = [sign] , unsigned_value;	value = sign__optional, unsigned_value;	value → sign__optional unsigned_value	value(1: !"0", !"1", !"2", !"3", !"4", !"5", !"6", !"7", !"8", !"9", !"+", !"-") → sign__optional unsigned_value	value = -sign >> unsigned_value >> BOUNDARIES;	value = sign__optional >> unsigned_value >> BOUNDARIES;	{LA_IS, { "unsigned_value_terminal", T_ADD_0, T_SUB_0 }, { "value", {\ {LA_IS, {""}, 2, { "sign__optional", "unsigned_value" }\ }}}\
	sign__optional = sign ε;	sign__optional → sign sign__optional → ε	sign__optional(1: !"+", !"-") → sign sign__optional(1: !"+", !"-") → ε		sign__optional = sign "";	{LA_IS, { T_ADD_0, T_SUB_0 }, { "sign__optional", {\ {LA_IS, {""}, 1, { "sign" } }\ }}}\ {LA_NOT, { T_ADD_0, T_SUB_0 }, { "sign__optional", {\ {LA_IS, {""}, 0, { "" } }\ }}}\
sign = sign_plus sign_minus;	sign = sign_plus sign_minus;	sign → sign_plus sign → sign_minus	sign(1: "+") → sign_plus sign(1: "-") → sign_minus	sign = sign_plus sign_minus;	sign = sign_plus sign_minus;	{LA_IS, { T_ADD_0 }, { "sign", {\ {LA_IS, {""}, 1, { "sign_plus" } }\ }}}\ {LA_IS, { T_SUB_0 }, { "sign", {\ {LA_IS, {""}, 1, { "sign_minus" } }\ }}}\
sign_plus = "+";	sign_plus = "+";	sign_plus → "+"	sign_plus(1: "+") → "+"	sign_plus = SAME_RULE(tokenPLUS);	sign_plus = SAME_RULE(tokenPLUS);	{LA_IS, { T_ADD_0 }, { "sign_plus", {\ {LA_IS, {""}, 1, {T_ADD_0} }\ }}}\
sign_minus = "-";	sign_minus = "-";	sign_minus → "-"	sign_minus(1: "-") → "-"	sign_minus = SAME_RULE(tokenMINUS);	sign_minus = SAME_RULE(tokenMINUS);	{LA_IS, { T_SUB_0 }, { "sign_minus", {\ {LA_IS, {""}, 1, {T_SUB_0} }\ }}}\
unsigned_value = non_zero_digit , {digit} "0";	unsigned_value = non_zero_digit , digit__iteration "0";	unsigned_value → non_zero_digit digit__iteration unsigned_value → "0"	unsigned_value(1: !"1", !"2", !"3", !"4", !"5", !"6", !"7", !"8", !"9") → non_zero_digit digit__iteration unsigned_value(1: !"0") → "0"	unsigned_value = (non_zero_digit >> *digit digit_0) >> BOUNDARIES;	unsigned_value = (non_zero_digit >> digit__iteration digit_0) >> BOUNDARIES;	/* unsigned_value_token represents unsigned_value in lexical analyzer */\ {LA_IS, { "unsigned_value_terminal", {\ {LA_IS, {""}, 1, { "unsigned_value_terminal" } }\ }}}\
	digit__iteration = digit, digit__iteration ε;	digit__iteration → digit digit__iteration digit__iteration → ε	digit__iteration(1: !"0", !"1", !"2", !"3", !"4", !"5", !"6", !"7", !"8", !"9") → digit digit__iteration digit__iteration(1: !"0", !"1", !"2", !"3", !"4", !"5", !"6", !"7", !"8", !"9") → ε		digit__iteration = digit >> digit__iteration "";	\
digit = "0" non_zero_digit;	digit = "0" non_zero_digit;	digit → "0" digit → non_zero_digit	digit(1: !"0") → "0" digit(1: !"1", !"2", !"3", !"4", !"5", !"6", !"7", !"8", !"9",) → non_zero_digit	digit_0 = '0'; digit = digit_0 non_zero_digit;	digit_0 = '0'; digit = digit_0 non_zero_digit;	\
non_zero_digit = "1" "2" "3" "4" "5" "6" "7" "8" "9";	non_zero_digit = "1" "2" "3" "4" "5" "6" "7" "8" "9";	non_zero_digit → "1" non_zero_digit → "2" non_zero_digit → "3" non_zero_digit → "4" non_zero_digit → "5" non_zero_digit → "6" non_zero_digit → "7" non_zero_digit → "8" non_zero_digit → "9"	non_zero_digit(1: !"1") → "1" non_zero_digit(1: !"2") → "2" non_zero_digit(1: !"3") → "3" non_zero_digit(1: !"4") → "4" non_zero_digit(1: !"5") → "5" non_zero_digit(1: !"6") → "6" non_zero_digit(1: !"7") → "7" non_zero_digit(1: !"8") → "8" non_zero_digit(1: !"9") → "9"	digit_1 = '1'; digit_2 = '2'; digit_3 = '3'; digit_4 = '4'; digit_5 = '5'; digit_6 = '6'; digit_7 = '7'; digit_8 = '8'; digit_9 = '9'; non_zero_digit = digit_1 digit_2 digit_3 digit_4 digit_5 digit_6 digit_7 digit_8 digit_9;	digit_1 = '1'; digit_2 = '2'; digit_3 = '3'; digit_4 = '4'; digit_5 = '5'; digit_6 = '6'; digit_7 = '7'; digit_8 = '8'; digit_9 = '9'; non_zero_digit = digit_1 digit_2 digit_3 digit_4 digit_5 digit_6 digit_7 digit_8 digit_9;	\
ident = "_" , letter_in_upper_case , letter_in_upper_case , letter_in_upper_case , letter_in_upper_case , letter_in_upper_case , letter_in_upper_case , letter_in_upper_case ;	ident = "_" , letter_in_upper_case , letter_in_upper_case , letter_in_upper_case , letter_in_upper_case , letter_in_upper_case , letter_in_upper_case , letter_in_upper_case ;	ident → "_" letter_in_upper_case letter_in_upper_case letter_in_upper_case letter_in_upper_case letter_in_upper_case letter_in_upper_case letter_in_upper_case	Ident(1: !"_) → "_" letter_in_upper_case letter_in_upper_case letter_in_upper_case letter_in_upper_case letter_in_upper_case letter_in_upper_case letter_in_upper_case	tokenUNDERSCORE = "_"; ident = { tokenINTEGER16 tokenCOMMA tokenNOT tokenAND tokenOR tokenEQUAL tokenNOTEQUAL tokenLESS	tokenUNDERSCORE = "_"; ident = { tokenINTEGER16 tokenCOMMA tokenNOT tokenAND tokenOR tokenEQUAL tokenNOTEQUAL tokenLESS	/* ident_token represents ident in lexical analyzer */\ {LA_IS, { "ident_terminal" }, { "ident", {\ {LA_IS, {""}, 1, {"ident_terminal" } }\ }}}\

				tokenGREATER tokenPLUS tokenMINUS tokenMUL tokenDIV tokenMOD tokenGROUPEXPRESSSIONBEGIN tokenGROUPEXPRESSSIONEND tokenLRASSIGN tokenELSE tokenIF tokenWHILE tokenCONTINUE tokenBREAK tokenEXIT tokenGET tokenPUT tokenNAME tokenBODY tokenDATA tokenBEGIN tokenEND tokenBEGINBLOCK tokenENDBLOCK tokenLEFTSQUAREBRACKETS tokenRIGHTSQUAREBRACKETS tokenSEMICOLON) >> tokenUNDERSCORE >> letter_in_upper_case >> letter_in_upper_case >> letter_in_upper_case >> letter_in_upper_case >> letter_in_upper_case >> letter_in_upper_case >> letter_in_upper_case >> STRICT_BOUNDARIES;	tokenGREATER tokenPLUS tokenMINUS tokenMUL tokenDIV tokenMOD tokenGROUPEXPRESSSIONBEGIN tokenGROUPEXPRESSSIONEND tokenLRASSIGN tokenELSE tokenIF tokenWHILE tokenCONTINUE tokenBREAK tokenEXIT tokenGET tokenPUT tokenNAME tokenBODY tokenDATA tokenBEGIN tokenEND tokenBEGINBLOCK tokenENDBLOCK tokenLEFTSQUAREBRACKETS tokenRIGHTSQUAREBRACKETS tokenSEMICOLON) >> tokenUNDERSCORE >> letter_in_upper_case >> letter_in_upper_case >> letter_in_upper_case >> letter_in_upper_case >> letter_in_upper_case >> letter_in_upper_case >> letter_in_upper_case >> STRICT_BOUNDARIES;	
letter_in_lower_case = "a" "b" "c" "d" "e" "f" "g" "h" "i" "j" "k" "l" "m" "n" "o" "p" "q" "r" "s" "t" "u" "v" "w" "x" "y" "z";	letter_in_lower_case → "a" letter_in_lower_case → "b" letter_in_lower_case → "c" letter_in_lower_case → "d" letter_in_lower_case → "e" letter_in_lower_case → "f" letter_in_lower_case → "g" letter_in_lower_case → "h" letter_in_lower_case → "i" letter_in_lower_case → "j" letter_in_lower_case → "k" letter_in_lower_case → "l" letter_in_lower_case → "m" letter_in_lower_case → "n" letter_in_lower_case → "o" letter_in_lower_case → "p" letter_in_lower_case → "q" letter_in_lower_case → "r" letter_in_lower_case → "s" letter_in_lower_case → "t" letter_in_lower_case → "u" letter_in_lower_case → "v" letter_in_lower_case → "w" letter_in_lower_case → "x" letter_in_lower_case → "y" letter_in_lower_case → "z"	letter_in_lower_case(1: "a") → "a" letter_in_lower_case(1: "b") → "b" letter_in_lower_case(1: "c") → "c" letter_in_lower_case(1: "d") → "d" letter_in_lower_case(1: "e") → "e" letter_in_lower_case(1: "f") → "f" letter_in_lower_case(1: "g") → "g" letter_in_lower_case(1: "h") → "h" letter_in_lower_case(1: "i") → "i" letter_in_lower_case(1: "j") → "j" letter_in_lower_case(1: "k") → "k" letter_in_lower_case(1: "l") → "l" letter_in_lower_case(1: "m") → "m" letter_in_lower_case(1: "n") → "n" letter_in_lower_case(1: "o") → "o" letter_in_lower_case(1: "p") → "p" letter_in_lower_case(1: "q") → "q" letter_in_lower_case(1: "r") → "r" letter_in_lower_case(1: "s") → "s" letter_in_lower_case(1: "t") → "t" letter_in_lower_case(1: "u") → "u" letter_in_lower_case(1: "v") → "v" letter_in_lower_case(1: "w") → "w" letter_in_lower_case(1: "x") → "x" letter_in_lower_case(1: "y") → "y" letter_in_lower_case(1: "z") → "z"	A = "A"; B = "B"; C = "C"; D = "D"; E = "E"; F = "F"; G = "G"; H = "H"; I = "I"; J = "J"; K = "K"; L = "L"; M = "M"; N = "N"; O = "O"; P = "P"; Q = "Q"; R = "R"; S = "S"; T = "T"; U = "U"; V = "V"; W = "W"; X = "X"; Y = "Y"; Z = "Z"; letter_in_lower_case = a b c d e f g h i j k l m n o p q r s t u v w x y z;	A = "A"; B = "B"; C = "C"; D = "D"; E = "E"; F = "F"; G = "G"; H = "H"; I = "I"; J = "J"; K = "K"; L = "L"; M = "M"; N = "N"; O = "O"; P = "P"; Q = "Q"; R = "R"; S = "S"; T = "T"; U = "U"; V = "V"; W = "W"; X = "X"; Y = "Y"; Z = "Z"; letter_in_lower_case = a b c d e f g h i j k l m n o p q r s t u v w x y z;	\	
letter_in_upper_case = "A" "B" "C" "D" "E" "F" "G" "H" "I" "J" "K" "L" "M" "N" "O" "P" "Q" "R" "S" "T" "U" "V" "W" "X" "Y" "Z";	letter_in_upper_case → "A" letter_in_upper_case → "B" letter_in_upper_case → "C" letter_in_upper_case → "D" letter_in_upper_case → "E" letter_in_upper_case → "F" letter_in_upper_case → "G" letter_in_upper_case → "H" letter_in_upper_case → "I" letter_in_upper_case → "J" letter_in_upper_case → "K" letter_in_upper_case → "L" letter_in_upper_case → "M" letter_in_upper_case → "N" letter_in_upper_case → "O" letter_in_upper_case → "P" letter_in_upper_case → "Q" letter_in_upper_case → "R" letter_in_upper_case → "S" letter_in_upper_case → "T" letter_in_upper_case → "U" letter_in_upper_case → "V" letter_in_upper_case → "W" letter_in_upper_case → "X" letter_in_upper_case → "Y" letter_in_upper_case → "Z"	letter_in_upper_case(1: "A") → "A" letter_in_upper_case(1: "B") → "B" letter_in_upper_case(1: "C") → "C" letter_in_upper_case(1: "D") → "D" letter_in_upper_case(1: "E") → "E" letter_in_upper_case(1: "F") → "F" letter_in_upper_case(1: "G") → "G" letter_in_upper_case(1: "H") → "H" letter_in_upper_case(1: "I") → "I" letter_in_upper_case(1: "J") → "J" letter_in_upper_case(1: "K") → "K" letter_in_upper_case(1: "L") → "L" letter_in_upper_case(1: "M") → "M" letter_in_upper_case(1: "N") → "N" letter_in_upper_case(1: "O") → "O" letter_in_upper_case(1: "P") → "P" letter_in_upper_case(1: "Q") → "Q" letter_in_upper_case(1: "R") → "R" letter_in_upper_case(1: "S") → "S" letter_in_upper_case(1: "T") → "T" letter_in_upper_case(1: "U") → "U" letter_in_upper_case(1: "V") → "V" letter_in_upper_case(1: "W") → "W" letter_in_upper_case(1: "X") → "X" letter_in_upper_case(1: "Y") → "Y" letter_in_upper_case(1: "Z") → "Z"	a = "a"; b = "b"; c = "c"; d = "d"; e = "e"; f = "f"; g = "g"; h = "h"; i = "i"; j = "j"; k = "k"; l = "l"; m = "m"; n = "n"; o = "o"; p = "p"; q = "q"; r = "r"; s = "s"; t = "t"; u = "u"; v = "v"; w = "w"; x = "x"; y = "y"; z = "z"; letter_in_upper_case = A B C D E F G H I J K L M N O P Q R S T U V W X Y Z;	a = "a"; b = "b"; c = "c"; d = "d"; e = "e"; f = "f"; g = "g"; h = "h"; i = "i"; j = "j"; k = "k"; l = "l"; m = "m"; n = "n"; o = "o"; p = "p"; q = "q"; r = "r"; s = "s"; t = "t"; u = "u"; v = "v"; w = "w"; x = "x"; y = "y"; z = "z"; letter_in_upper_case = A B C D E F G H I J K L M N O P Q R S T U V W X Y Z;	\	
				STRICT_BOUNDARIES = (BOUNDARY >> *(BOUNDARY)) ((qi::alpha qi::char("_"));); BOUNDARIES = (BOUNDARY >> *(BOUNDARY) NO_BOUNDARY); BOUNDARY = BOUNDARY__SPACE BOUNDARY__TAB BOUNDARY__VERTICAL_TAB BOUNDARY__FORM_FEED BOUNDARY__CARRIAGE_RETURN BOUNDARY__LINE_FEED BOUNDARY__NULL; BOUNDARY__SPACE = " "; BOUNDARY__TAB = "\t"; BOUNDARY__VERTICAL_TAB = "\v"; BOUNDARY__FORM_FEED = "\f"; BOUNDARY__CARRIAGE_RETURN = "\r"; BOUNDARY__LINE_FEED = "\n"; BOUNDARY__NULL = "\0"; NO_BOUNDARY = ""; #define WHITESPACES \ STRICT_BOUNDARIES, \	STRICT_BOUNDARIES = (BOUNDARY >> *(BOUNDARY)) ((qi::alpha qi::char("_"));); BOUNDARIES = (BOUNDARY >> *(BOUNDARY) NO_BOUNDARY); BOUNDARY = BOUNDARY__SPACE BOUNDARY__TAB BOUNDARY__VERTICAL_TAB BOUNDARY__FORM_FEED BOUNDARY__CARRIAGE_RETURN BOUNDARY__LINE_FEED BOUNDARY__NULL; BOUNDARY__SPACE = " "; BOUNDARY__TAB = "\t"; BOUNDARY__VERTICAL_TAB = "\v"; BOUNDARY__FORM_FEED = "\f"; BOUNDARY__CARRIAGE_RETURN = "\r"; BOUNDARY__LINE_FEED = "\n"; BOUNDARY__NULL = "\0"; NO_BOUNDARY = ""; #define WHITESPACES \	\

				BOUNDARIES, \ BOUNDARY, \ BOUNDARY__SPACE, \ BOUNDARY__TAB, \ BOUNDARY__VERTICAL_TAB, \ BOUNDARY__FORM_FEED, \ BOUNDARY__CARRIAGE_RETURN, \ BOUNDARY__LINE_FEED, \ BOUNDARY__NULL, \ NO_BOUNDARY	STRICT_BOUNDARIES, \ BOUNDARIES, \ BOUNDARY, \ BOUNDARY__SPACE, \ BOUNDARY__TAB, \ BOUNDARY__VERTICAL_TAB, \ BOUNDARY__FORM_FEED, \ BOUNDARY__CARRIAGE_RETURN, \ BOUNDARY__LINE_FEED, \ BOUNDARY__NULL, \ NO_BOUNDARY	
						\ \ \ \ { LA_IS, { T_NAME_0 }, { "program____part1", { \ { LA_IS, { "" }, 7, { T_NAME_0, "program_name", \ T_SEMICOLON_0, T_BODY_0, T_DATA_0, \ "declaration_optional", T_SEMICOLON_0 }} \ }}} \ \ } \ "program_rule"